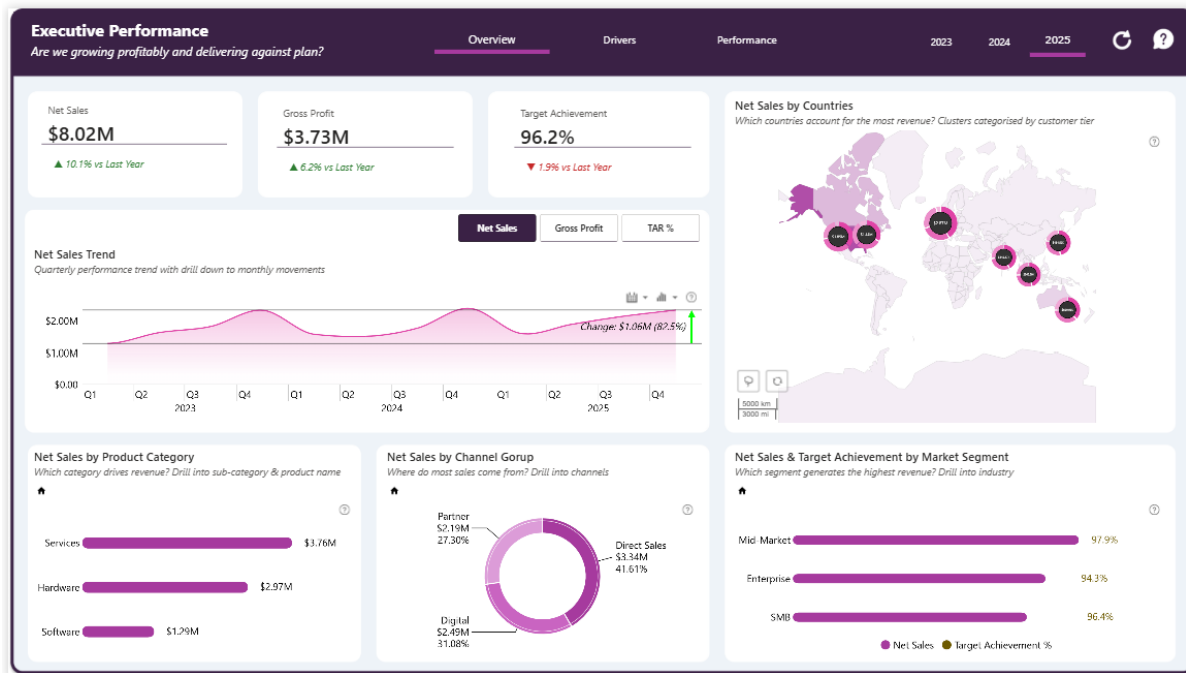


Global B2B Sales Performance & Profitability Analytics

Executive Power BI portfolio case study

Financial performance • strategic growth • customer and product mix • commercial effectiveness • revenue leakage



Final Executive Performance page

Power BI

DAX

Power Query

ZoomCharts

Star schema

Audience: senior partners, executive leadership and commercial management

Dataset: 6,000 sales lines • 3,828 orders • 900 customers • 28 products • 36 salespeople • 2023–2025

Executive summary

A decision-focused B2B sales dashboard designed for senior management.

Project purpose

Transform a multi-domain global B2B sales dataset into a three-page executive report that helps leaders monitor financial health, understand the commercial drivers behind growth, and diagnose where discounts, returns, fulfilment costs and exceptional business events erode value.

What I delivered

- A three-page Power BI report covering Executive Performance, Commercial Drivers, and Commercial Effectiveness & Risk, connected through consistent navigation, filters and drill-down behaviour.
- A star-schema semantic model linking customer, product, geography, salesperson, channel, campaign, business-event and date dimensions to a central sales transaction table.
- A reusable DAX measure layer for net sales, gross profit, margin, targets, returns, discount cost, strategic revenue streams, business-event exposure, shipping and time comparisons.
- Disconnected selectors and dynamic measures to switch strategic growth metrics, update chart titles and tooltip labels, and reduce visual duplication.
- A management narrative that separates headline performance from strategic growth mix and controllable commercial leakage.

Headline dashboard results — 2025 view

Decision area	Dashboard result
Financial performance	\$8.02M Net Sales and \$3.73M Gross Profit; Net Sales grew 10.1% YoY while Gross Profit grew 6.2%.
Target delivery	96.2% target achievement, leaving a \$313.5K shortfall to the 2025 sales target.
Strategic growth mix	\$1.69M Self-Service Revenue (+14.4% YoY) and \$2.24M Premium Portfolio Gross Profit (+15.1% YoY), while Recurring and New Launch Revenue declined.
Commercial leakage	\$326.8K discount cost, \$381.0K returned sales and \$49.3K shipping cost in the selected year.
Business-event exposure	\$778.5K of 2025 Net Sales is linked to defined business events; Holiday Hardware Surge is the largest event contribution.

Portfolio summary

Built a three-page Power BI dashboard for a global B2B sales challenge covering financial performance, strategic customer and product growth, routes to market, promotions, returns and business events. Developed an inferred data dictionary, star-schema model, reusable DAX measures, dynamic selectors and ZoomCharts drill-down visuals to reveal target gaps, growth-quality trade-offs and sources of commercial leakage. The report enables senior leaders to move from outcome to driver to action without turning each source table into a separate dashboard page.

1. Business context and objectives

From transaction data to executive commercial decisions.

The challenge required an interactive B2B sales dashboard that helps business leaders understand revenue performance, profitability, target achievement, customer behaviour and the factors driving results across global markets. The analytical challenge was not simply to rank dimensions by sales, but to organise a wide dataset around the decisions senior leaders make: whether growth is profitable, whether the growth mix is strategically healthy, and where commercial value is being lost.

Primary stakeholders

Stakeholder	Information need	Decision supported
Senior partners / executives	Net Sales, Gross Profit, target delivery, growth quality and major risk signals.	Investment priorities, performance escalation and management focus.
Sales leadership	Territory performance, discount leakage, customer/order behaviour and target gaps.	Coaching, pricing discipline, account strategy and resource allocation.
Finance / commercial finance	Margin, returns, discount cost, shipping cost and target variance.	Margin protection, commercial controls and value-leakage review.
Product / portfolio leaders	Product lifecycle, premium portfolio value, new-launch contribution and return hotspots.	Portfolio investment, lifecycle actions and product remediation.
Marketing / channel leaders	Promotion efficiency, self-service adoption and route-to-market mix.	Campaign governance, channel strategy and digital adoption.

Success criteria

- Answer the core executive questions within one or two interactions rather than requiring detailed transaction reports.
- Keep the three report pages analytically distinct: performance, drivers, then effectiveness and risk.
- Respect sales-line versus order-level grain so order counts, cancellation and shipping measures are not inflated by multi-line orders.
- Use drill-down to move from high-level categories into product, customer, territory, salesperson and time detail without overcrowding the first view.
- Maintain a consistent executive design with concise titles, contextual KPI references, limited colours, help icons and reset controls.

North-star question

Management question

Are we growing profitably and meeting plan through the right customers, products and routes to market — without giving away value through discounts, returns, fulfilment costs or exceptional trading conditions?

2. Dataset and analytical scope

A global B2B sales model spanning financial, customer, product, commercial and event dimensions.

The supplied challenge workbook contains 10 worksheets: eight business dimensions, one central sales fact table and a sheet of suggested DAX measures. The source covers 6,000 sales lines from 2023 to 2025 across North America, Europe and Asia Pacific.

Core field groups

Field group	Examples
Sales transaction	Order ID, order/ship date keys, quantity, list price, discount %, sales amount, COGS, Gross Profit, shipping cost, returned amount and target sales amount.
Customers	Customer segment, industry, customer name, lifecycle stage, customer tier, account status, acquisition year and estimated annual revenue.
Products	Category, subcategory, product, recurring-revenue flag, portfolio tier, business unit, brand, lifecycle stage and launch year.
Sales organisation	Region, sales team, manager, salesperson, role, sales director and territory.
Channels	Channel group, channel and Assisted / Self-service motion.
Marketing campaigns	Promotion name/type, default discount, campaign, discount strategy, season and marketing objective.
Business events	Event type, affected region, start/end dates, business explanation and expected impact.
Geography and time	Region, country, state/province, city, currency, coordinates, year, quarter, month and calendar attributes.

Key analytical questions

Report page	Questions answered
Executive Performance	Are Net Sales and Gross Profit growing? Are we meeting target? Which countries, product categories, channels and market segments drive the result?
Commercial Drivers	Which strategic customer, product and route-to-market levers are strengthening or weakening? Is growth coming from recurring, new-launch, self-service and premium business?
Commercial Effectiveness & Risk	Where are discounts, returns, shipping and business events affecting value? Which promotions, territories and product areas require intervention?

3. Data preparation and semantic modelling

Building reliable business definitions before building visuals.

Data preparation approach

- Reverse-engineered the workbook because no official data dictionary was provided; documented 107 fields, table grain, keys, observed domains, relationships and analytical caveats.
- Validated the 6,000-row Sales Transactions table as sales-line grain and identified 3,828 distinct orders.
- Confirmed primary keys were unique and fact foreign keys matched the related dimensions; no material null-value problem was identified in the supplied workbook.
- Renamed model tables and fields for business readability, including Customer, Product, Location, Account Managers, Marketing Campaign, Business Event and Sales Transactions.
- Separated reusable calculations into Core Measures, Time Comparison Measures and Dynamic Measures tables.
- Created disconnected selector/parameter tables for strategic-growth switching, overview metric selection and waterfall sequencing.

Key quality and modelling issues

Issue	Treatment
Sales-line vs order grain	Order Count uses DISTINCTCOUNT(Order ID); shipping-duration logic iterates distinct orders so multi-line orders are not overweighted.
Promotion grain	659 orders contain more than one promotion key, so campaign analysis is treated at sales-line level rather than assuming one campaign per order.
Sales terminology	Sales Amount is already net of discount; the supplied “Gross Sales” measure therefore represents sales after discount but before returns.
Profit reconciliation	Verified that Gross Profit reconciles to Sales Amount – Returned Amount – COGS – Shipping Cost across the source records.
Currency ambiguity	ExchangeRateToUSD exists, but the supplied KPI logic does not apply it. Currency conversion was not imposed without a documented semantic rule.
Identifier quality	Salesperson names are not unique; SalespersonKey is retained as the stable identifier for model relationships and detailed analysis.

Figure 1. Power BI semantic model

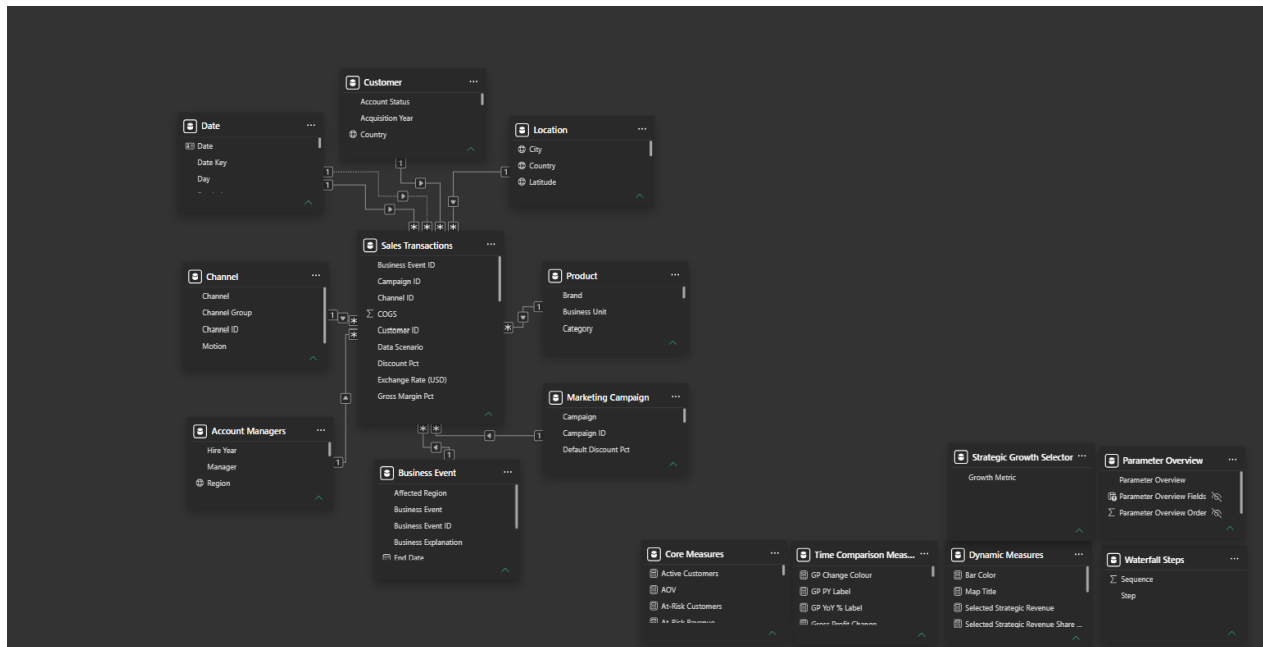


Figure 1. Power BI semantic model

The final model centres Sales Transactions between conformed business dimensions. Disconnected selector and parameter tables sit outside the star schema, while dedicated measure tables keep reusable business logic separate from source attributes.

4. Measure design and report interaction

DAX definitions were aligned to business meaning, source grain and executive interpretation.

Core calculation principles

Measure	Definition	Reason
Net Sales	Sales Amount – Returned Sales	Measures retained revenue after product returns.
Gross Margin %	Gross Profit ÷ Net Sales	Uses weighted aggregate margin rather than averaging row-level percentages.
Target Achievement %	Net Sales ÷ Sales Target	Provides a consistent plan-delivery measure across time and dimensions.
Effective Discount Rate	Discount Cost ÷ List Sales	Weights discounting by the commercial value of each transaction instead of averaging line discount percentages.
Return Rate %	Returned Sales ÷ Gross Sales	Quantifies the share of recognised sales value subsequently returned.
Average Order-to-Ship Days	Average date difference across distinct non-cancelled orders	Prevents orders with many sales lines from receiving extra weight.

Interaction architecture

- Top navigation moves between Overview, Drivers and Performance sections while preserving a consistent report shell.
- Year buttons support 2023–2025 comparison and drive KPI prior-year / YoY context.
- The Executive Performance trend uses a selector for Net Sales, Gross Profit and Target Achievement without duplicating the chart.
- The Commercial Drivers trend uses a disconnected Strategic Growth Selector to switch Recurring Revenue, New Launch Revenue and Self-Service Revenue together with the matching share-of-Net-Sales line.
- ZoomCharts drill-down paths expose subcategory/product, customer/industry, territory/team/salesperson and time detail only when required.
- Help icons, concise subtitles and reset controls make the calculation scope and interaction path discoverable.

Example: effective discount rate

Effective Discount Rate = DIVIDE ([Discount Cost \$], [List Sales])

This measure answers the management question “What percentage of potential list-price revenue was given away as discount?” and is used as the contextual reference beneath Discount Cost on the risk page.

5. Report design strategy

A restrained executive design focused on analytical questions, not chart decoration.

Design system

Design element	Implementation
Navigation	Three tab-style buttons: Overview, Drivers and Performance, with the current page highlighted.
KPI layout	Three headline cards on the executive overview; five contextual KPI cards on each diagnostic page.
Visual density	Five principal visuals per page, sized to preserve whitespace and keep the main question readable.
Colour	Deep plum for navigation and structural emphasis, magenta/purple for primary analytical series, neutral report surfaces, and green/red only for favourable/unfavourable KPI movement.
Titles	Business-question titles such as “Promotion Efficiency: Discount vs Margin” rather than field-name labels.
Subtitles	Short prompts describe the decision question or drill path expected from each visual.
Drill-down	ZoomCharts visuals provide hierarchical navigation without displaying every customer, product or salesperson at once.
Help and reset	Information icons explain interpretation; the reset control restores the default report state after exploration.

Why the report uses three pages

Page	Management lens	Executive decision
Executive Performance	Financial outcome and plan delivery	How are we performing, and where is the result concentrated?
Commercial Drivers	Strategic customer, product and route-to-market mix	What kind of growth are we building?
Commercial Effectiveness & Risk	Commercial actions, leakage and exceptional conditions	Where are we losing value, and why?

Design principle

The report moves from outcome, to growth mix, to leakage and intervention. This prevents Page 2 and Page 3 from becoming repeated “Net Sales by dimension” views and gives senior leaders a clear progression from “what happened?” to “what should we investigate?”

6. Page 1 — Executive Performance

Headline sales, profitability, target delivery and market concentration.

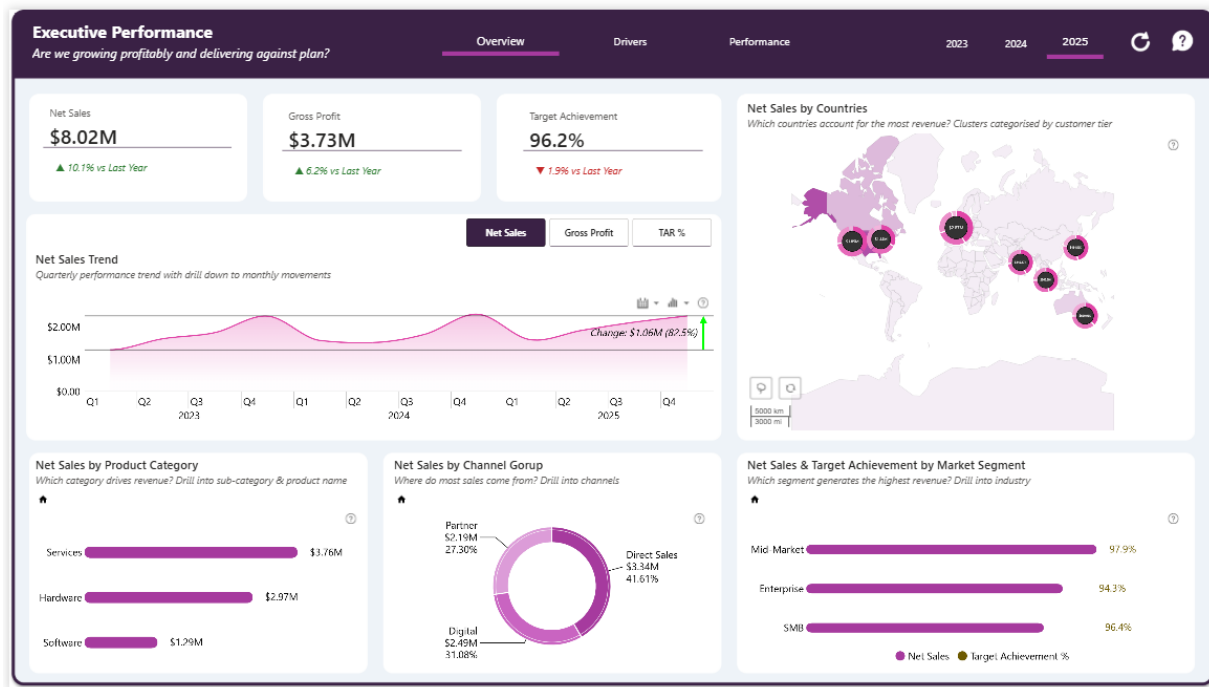


Figure 2. Executive Performance page

Page objective

Give senior leaders an immediate view of financial health and plan delivery, then show where the result is concentrated across time, countries, products, channels and customer segments. The page deliberately limits the KPI strip to three headline outcomes so the trend and map remain prominent.

Key visuals and decisions

Visual	Question answered	Management use
Dynamic performance trend	How are Net Sales, Gross Profit or Target Achievement changing over time?	Investigate periods where growth, profitability or plan delivery changes direction.
Country map	Which countries contribute most to Net Sales?	Identify geographic concentration and prioritise market-level review.
Product category ranking	Which product categories drive revenue?	Drill into subcategory and product to understand scale and mix.
Channel group mix	Where do sales come from across Direct, Digital and Partner routes?	Assess route-to-market dependence and channel contribution.
Market segment value & target	Which customer segments combine scale with target delivery?	Compare Enterprise, Mid-Market and SMB performance.

Observed Executive Performance insights — 2025

- Net Sales reached \$8.02M, up 10.1% versus 2024, while Gross Profit reached \$3.73M, up 6.2%.
- Gross Margin was 46.5%, down from 48.2% in 2024, showing that profit growth did not keep pace with revenue growth.
- Target Achievement was 96.2%, leaving a \$313.5K shortfall and declining by approximately 1.9 percentage points from 2024.
- Direct Sales was the largest channel at \$3.34M (41.6% of Net Sales), followed by Digital at \$2.49M and Partner at \$2.19M.
- Services was the largest product category at \$3.76M, while Software generated only \$1.29M of Net Sales but carried a materially higher 77.8% Gross Margin.
- Mid-Market generated the most Net Sales at \$2.97M and achieved 97.9% of target; Enterprise generated \$2.62M but had the weakest segment target achievement at 94.3%.
- The United States was the largest country market at approximately \$2.29M of Net Sales.

Management implications

- Treat 2025 as strong top-line growth with weaker quality: revenue accelerated, but margin rate and target delivery both deteriorated.
- Investigate the Enterprise target gap rather than assuming its sizeable revenue base translates into healthy plan performance.
- Protect the economics of Software and assess whether its high-margin contribution can be expanded without compromising customer fit.
- Use channel and country drill-downs to distinguish structural growth from concentration in a small number of routes or markets.
- Carry the target gap forward to Page 3, where discounting, returns, business events and shipping costs are diagnosed as potential contributors.

7. Page 2 — Commercial Drivers

Strategic customer, product and route-to-market growth.

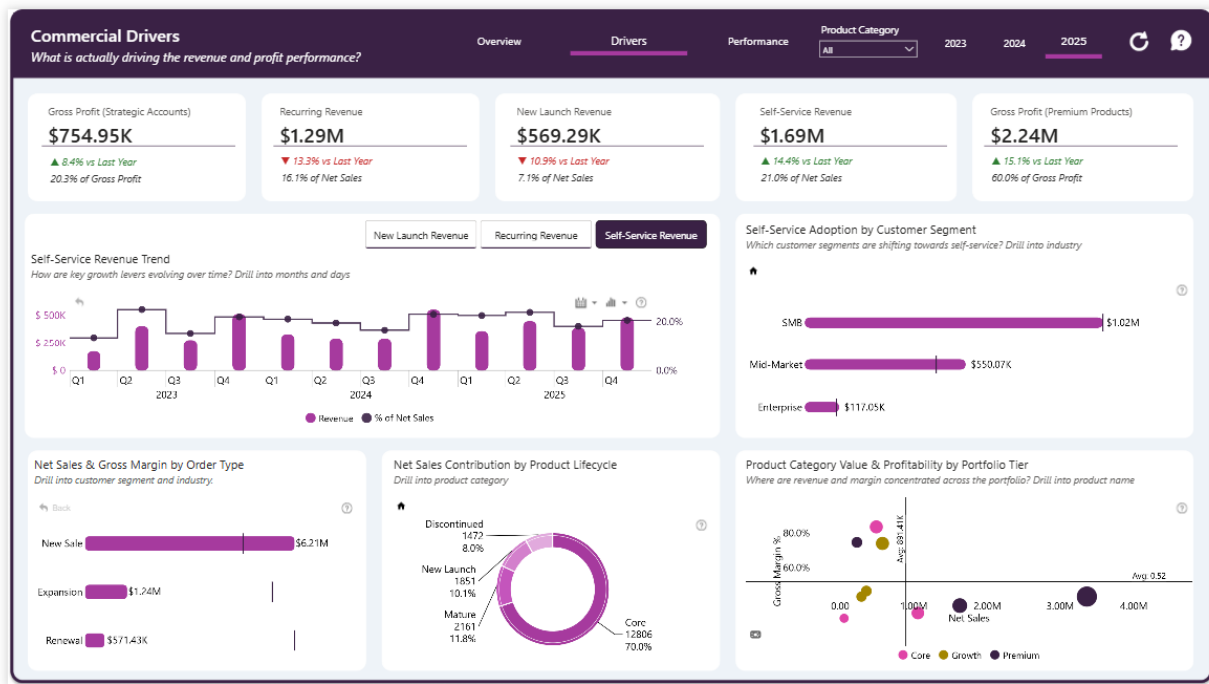


Figure 3. Commercial Drivers page

Page objective

Explain what kind of business is driving the headline result. The page focuses on strategic account value, recurring-product revenue, new-launch adoption, self-service growth and premium portfolio contribution rather than repeating the financial overview.

Key visuals and decisions

Visual	Question answered	Management use
Strategic growth trend	How are Recurring, New Launch and Self-Service Revenue evolving, and what share of Net Sales do they represent?	Track whether strategic growth levers are strengthening or weakening over time.
Self-Service adoption by segment	Which customer segments are adopting self-service?	Identify where digital adoption is strong and where assisted selling remains dominant.
Order type value & margin	How do New Sale, Expansion and Renewal differ in scale and profitability?	Assess dependence on acquisition versus deeper existing-customer relationships.
Product lifecycle mix	How much business comes from Core, New Launch, Mature and Discontinued products?	Monitor portfolio health and ageing exposure.
Portfolio tier value & profitability	Which portfolio tiers combine Net Sales scale with attractive Gross Margin?	Balance portfolio growth, premium value and margin quality.

Observed Commercial Drivers insights — 2025

- Strategic-account Gross Profit reached \$754.9K, contributing 20.3% of total Gross Profit and increasing 8.4% YoY.
- Self-Service Revenue reached \$1.69M, representing 21.0% of Net Sales and growing 14.4% YoY.
- Self-service adoption is highly uneven: approximately 42.0% of SMB revenue is self-service, compared with 18.5% for Mid-Market and only 4.5% for Enterprise.
- Recurring Revenue fell 13.3% YoY to \$1.29M and New Launch Revenue fell 10.9% to \$569.3K, even though total Net Sales grew strongly.
- New Sale generated \$6.21M of Net Sales (77.4% of the total) at a 44.2% Gross Margin, while Expansion and Renewal were smaller but carried higher margins of 52.3% and 58.5% respectively.
- Premium products generated \$2.24M of Gross Profit — 60.0% of total Gross Profit — and grew 15.1% YoY. Growth-tier products were smaller by sales but carried a higher aggregate margin than Premium.
- Core products generated approximately 62% of 2025 Net Sales; Mature and Discontinued products together still accounted for roughly 31%, while New Launch contributed about 7%.

Management implications

- Do not interpret total revenue growth as uniformly healthy: two strategic growth levers — recurring and new-launch revenue — contracted in 2025.
- Expansion and Renewal activity appears economically attractive but under-scaled; investigate account-management opportunities to deepen existing relationships.
- Use self-service selectively. SMB adoption is already strong, while Enterprise customers remain predominantly assisted and may require a different migration strategy.
- Protect the Premium portfolio because it drives the majority of Gross Profit, but compare its margin economics with the higher-margin Growth tier before prioritising investment.
- Monitor mature/discontinued exposure and ensure portfolio lifecycle decisions do not undermine the recurring and new-launch growth agenda.

8. Page 3 — Commercial Effectiveness & Risk

Discounting, returns, exceptional events and commercial value leakage.

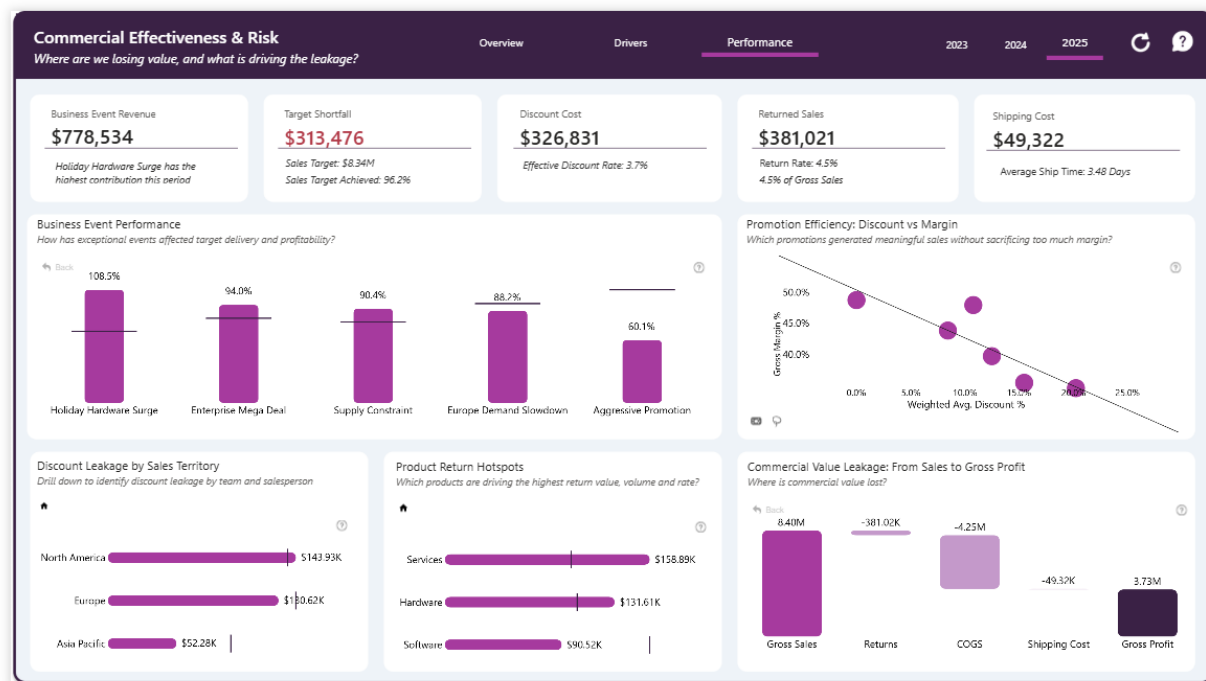


Figure 4. Commercial Effectiveness & Risk page

Page objective

Diagnose why strong commercial activity does not always translate into target-achieving, high-quality profit. The page connects exceptional trading events, promotion economics, discount behaviour, product returns and fulfilment costs to the final retained Gross Profit.

Key visuals and decisions

Visual	Question answered	Management use
Business Event Performance	How did each defined event perform against target and profitability expectations?	Separate unusual trading conditions from normal business performance.
Promotion efficiency	Which promotions generate meaningful sales without sacrificing too much margin?	Apply profitability guardrails to discount-led campaigns.
Discount leakage by territory	Where is discount value being surrendered across the sales organisation?	Drill into team and salesperson to target pricing review.
Product return hotspots	Which product categories drive the highest returned value and rate?	Prioritise product, service or fulfilment investigation.
Commercial value leakage	How does value move from Gross Sales to Gross Profit?	Quantify the cumulative impact of returns, COGS and shipping.

Observed Commercial Effectiveness & Risk insights

- In the 2025 view, the business carries a \$313.5K target shortfall, \$326.8K discount cost, \$381.0K returned sales and \$49.3K shipping cost.
- The 2025 Effective Discount Rate is 3.7% of list-price sales, while Return Rate is 4.5% of Gross Sales and average order-to-ship time is 3.48 days.
- Business Event Revenue totals \$778.5K in 2025, approximately 9.7% of Net Sales; Holiday Hardware Surge is the largest event contribution.
- Across the defined event periods, Holiday Hardware Surge achieved 108.5% of target but only a 32.7% Gross Margin, showing that target outperformance does not automatically mean high-quality growth.
- Aggressive Promotion was the weakest event by target achievement at approximately 60.1%; Europe Demand Slowdown and Supply Constraint achieved approximately 88.2% and 90.4% respectively.
- Promotion analysis shows a clear margin trade-off as effective discounts deepen: Holiday Campaign is around 15.5% discount / 35.5% margin and Strategic Deal around 20.2% discount / 34.6% margin.
- North America carries the largest 2025 discount cost at about \$143.9K, followed by Europe at \$130.6K and Asia Pacific at \$52.3K.
- Services has the highest returned-sales value at about \$158.9K, while Software has the highest category return rate at approximately 6.6%.
- The 2025 waterfall moves from \$8.40M Gross Sales to \$3.73M Gross Profit after \$381.0K returns, approximately \$4.25M COGS and \$49.3K shipping cost.

Management implications

- Judge event and promotion success on retained margin as well as target or revenue uplift; the Holiday surge illustrates why volume alone can overstate quality.
- Introduce pricing review thresholds for territories, teams or salespeople with high discount cost and high effective discount rates.
- Treat return value and return rate as separate signals: Services has the largest dollar exposure, while Software has the highest rate.
- Separate controllable leakage — discounting, returns and fulfilment — from external/event-driven performance before assigning accountability.
- Use the waterfall as the closing management view: it translates multiple operational pressures into the amount of Gross Profit ultimately retained.

9. Cross-page insights and recommendations

The report is designed to connect financial outcome, strategic mix and commercial leakage.

Integrated findings

Insight	Evidence	Management meaning
Top-line growth is stronger than profit quality	2025 Net Sales +10.1% YoY vs Gross Profit +6.2%; margin fell to 46.5% and target achievement to 96.2%.	Growth should be reviewed with margin, returns and target delivery rather than revenue alone.
Strategic growth is uneven	Self-Service Revenue +14.4% and Premium GP +15.1%, while Recurring Revenue –13.3% and New Launch Revenue –10.9%.	Headline growth is not strengthening every strategic growth engine.
Existing-customer economics are attractive but under-scaled	Expansion and Renewal together are a minority of Net Sales but carry materially higher margins than New Sale.	Account expansion and renewal deserve focused commercial-development attention.
Discounting and events can create low-quality growth	Deeper promotion discounts align with weaker margins; Holiday Hardware Surge beat target but at 32.7% margin.	Target attainment should not override profitability guardrails.
Leakage is multi-source	\$326.8K discount cost, \$381.0K returned sales and \$49.3K shipping cost in 2025.	Management action must address pricing, product/return issues and fulfilment rather than searching for one cause.
Customer/channel adoption differs materially	SMB self-service share is about 42% of segment revenue versus 4.5% for Enterprise.	Digital strategy should be segment-specific rather than uniform.

Recommended management actions

- Target recovery: isolate the Enterprise segment, countries and territories contributing most to the target gap and define account-level recovery actions.
- Growth mix: investigate the 2025 decline in recurring and new-launch revenue even while total sales are growing.
- Customer development: create expansion and renewal plays for existing accounts, where the current margin profile is stronger than New Sale.
- Promotion governance: set margin and discount guardrails by campaign objective; assess promotions against incremental value, not sales volume alone.
- Return reduction: investigate Services for returned-value exposure and Software for return-rate exposure, drilling to subcategory/product.
- Commercial discipline: review North American and European discount leakage by territory, team and salesperson.
- Event-adjusted management: compare exceptional-event periods with normal trading before attributing performance solely to sales execution.

Decision value

The report does not prescribe a single “good” benchmark for every market or campaign. It gives senior leaders a structured path to identify the exception, drill into the responsible customer/product/team/event, and assign the next management action.

10. Challenges, limitations and future enhancements

Transparent assumptions strengthen the credibility of the analysis.

Challenges solved during development

Challenge	Resolution
No data dictionary	Reverse-engineered the workbook and created an inferred field dictionary covering 107 fields, relationships, domains and caveats.
Mixed analytical grain	Kept sales calculations at line level while order metrics use distinct Order ID logic.
Page overlap	Separated overview financials, strategic drivers and commercial-risk diagnostics instead of repeating Net Sales across every page.
Dynamic visual requirements	Used disconnected selectors, SWITCH measures and dynamic titles so ZoomCharts receives ordinary DAX measures while the user can change the strategic metric.
Visual overcrowding	Used hierarchy drill-down, a three-KPI overview and focused five-visual detail pages.
Interpretation risk	Added subtitles, KPI reference labels, help icons and explicit business-question titles.

Limitations

- ExchangeRateToUSD is supplied but the intended reporting-currency conversion rule is not documented; the headline measures follow the supplied semantic logic rather than imposing an unverified conversion.
- RecurringRevenueFlag is a product classification and should not be interpreted as accounting ARR or MRR without contract/subscription data.
- Business-event tagging supports scenario comparison but does not establish statistical causality between an event and the observed sales outcome.
- Campaign objectives such as Pipeline Conversion cannot be measured directly because lead/opportunity pipeline data is not included; order and revenue outcomes can only act as proxies.
- Customer and product lifecycle fields are snapshot attributes, not historical slowly-changing dimensions; lifecycle transition analysis would require dated history.
- “New Sale” is an order type and does not necessarily mean a newly acquired customer.
- The supplied targets are transaction-level target values rather than a formal quota/forecast model by salesperson, product or market.

Future enhancements

- Add opportunity, marketing spend and campaign-cost data to calculate conversion, incremental lift and campaign ROI.
- Add subscription contracts, ARR/MRR, renewal dates and churn to deepen recurring-revenue analysis.
- Introduce inventory and fulfilment-service data to quantify the causal impact of supply constraints and shipping performance.
- Add customer cohorts, repeat-purchase and retention metrics across acquisition year and lifecycle.
- Create target/forecast tables at salesperson, product and region level with variance-to-plan ownership.
- Introduce row-level security, drill-through detail pages, deployment pipelines and automated data-quality monitoring for production use.

11. Skills demonstrated

A portfolio project combining technical delivery with executive decision design.

Capability	Evidence	Value
Power BI development	Three-page executive layout, navigation, slicers, KPI cards, conditional formatting, help icons and report interactions.	Built a decision interface rather than a collection of charts.
Data modelling	Star schema, role-aware date relationships, dimension keys, disconnected selector tables and dedicated measure tables.	Improved model reliability and simplified report logic.
DAX	Financial measures, target/YoY logic, distinct-order calculations, contribution measures, dynamic metric selection, titles and tooltip labels.	Created reusable, consistent business definitions.
Data quality	Grain validation, key checks, formula reconciliation, domain profiling and semantic caveat documentation.	Reduced the risk of misleading calculations and visuals.
Analytical storytelling	Outcome → Drivers → Risk page narrative, question-led titles and management implications.	Translated technical output into executive decisions.
UX / ZoomCharts	Drill-down timelines, maps, hierarchical bars, scatter analysis, dynamic selectors and waterfall storytelling.	Supported exploration without overwhelming senior users.

Recruiter-facing project description

Global B2B Sales Performance & Profitability Analytics

Designed and developed a three-page executive Power BI dashboard analysing 6,000 B2B sales lines across 2023–2025. Built an inferred data dictionary, star-schema semantic model, reusable DAX layer, dynamic selectors and ZoomCharts drill-downs to help senior leaders understand financial performance, strategic growth drivers, promotion and discount effectiveness, returns, business events and commercial value leakage.

60-second interview explanation

I built an executive B2B sales analytics report from a global dataset covering 6,000 sales lines across customers, products, channels, sales teams, promotions and business events. Because the workbook had no data dictionary, I first reverse-engineered the grain, keys and financial calculations and documented the semantic rules. I then structured the report into three management questions: how the business is performing, what is driving that performance, and where commercial value is being lost. The model uses a star schema, reusable DAX measures and disconnected selectors for dynamic ZoomCharts. The 2025 analysis showed strong revenue growth but weaker target and margin quality, strong self-service and premium-product growth, declines in recurring and new-launch revenue, and material leakage through discounting and returns. The main value of the project was turning a wide synthetic dataset into a coherent executive decision path rather than building one page per source table.

Three-line portfolio description

Built a three-page Power BI dashboard for global B2B sales performance, strategic growth and commercial risk. Developed a star-schema model, inferred data dictionary, reusable DAX measures and ZoomCharts drill-downs to analyse targets, customer/product mix, promotions, returns, business events and revenue leakage. The report helps senior leaders move from headline performance to underlying growth drivers and specific areas requiring commercial action.

Appendix A — Selected DAX measures

Representative calculations used in the semantic model.

Net Sales and profitability

```
Gross Sales =
SUM ( 'Sales Transactions'[Sales Amount] )

Returned Sales =
SUM ( 'Sales Transactions'[Returned Amount] )

Net Sales =
[Gross Sales] - [Returned Sales]

Gross Profit =
SUM ( 'Sales Transactions'[Gross Profit] )

Gross Margin % =
DIVIDE ( [Gross Profit], [Net Sales] )
```

Target and discounting

```
Sales Target =
SUM ( 'Sales Transactions'[Target Sales Amount] )

Target Achievement % =
DIVIDE ( [Net Sales], [Sales Target] )

Target Shortfall =
MAX ( 0, [Sales Target] - [Net Sales] )

List Sales =
SUMX ( 'Sales Transactions', 'Sales Transactions'[Quantity] * 'Sales Transactions'[List Unit Price] )

Discount Cost $ =
[List Sales] - [Gross Sales]

Effective Discount Rate =
DIVIDE ( [Discount Cost $], [List Sales] )
```

Appendix B — Dynamic and order-grain measures

Additional patterns used for strategic switching, tooltips and operational interpretation.

Dynamic strategic-growth selector

```
Strategic Growth Selector =
DATATABLE (
    "Growth Metric", STRING,
    {
        { "Recurring Revenue" },
        { "New Launch Revenue" },
        { "Self-Service Revenue" }
    }
)

Selected Strategic Revenue =
SWITCH (
    SELECTEDVALUE ( 'Strategic Growth Selector'[Growth Metric], "Recurring Revenue" ),
    "Recurring Revenue", [Recurring Revenue],
    "New Launch Revenue", [New Launch Revenue],
    "Self-Service Revenue", [Self-Service Revenue]
)

Selected Strategic Revenue Share % =
DIVIDE ( [Selected Strategic Revenue], [Net Sales] )

Strategic Growth Trend Title =
SELECTEDVALUE (
    'Strategic Growth Selector'[Growth Metric],
    "Recurring Revenue"
) & " Trend"
```

Distinct-order shipping logic

```
Avg. Order-to-Ship Days =
AVERAGEX (
    FILTER (
        VALUES ( 'Sales Transactions'[Order ID] ),
        CALCULATE ( MAX ( 'Sales Transactions'[Order Status] ) ) <> "Cancelled"
    ),
    VAR OrderDateKey = CALCULATE ( MIN ( 'Sales Transactions'[Order Date Key] ) )
    VAR ShipDateKey = CALCULATE ( MIN ( 'Sales Transactions'[Ship Date Key] ) )
    RETURN DATEDIFF (
        DATE ( INT ( OrderDateKey / 10000 ), INT ( MOD ( OrderDateKey, 10000 ) / 100 ), MOD ( OrderDateKey, 100 ) ),
        DATE ( INT ( ShipDateKey / 10000 ), INT ( MOD ( ShipDateKey, 10000 ) / 100 ), MOD ( ShipDateKey, 100 ) ),
        DAY
    )
)
```

Final project reflection

The strongest part of the project was not any individual visual. It was the decision to reverse-engineer the dataset first, define the sales-line and order-level logic explicitly, and then organise the report around three management questions. That prevented visually attractive but repetitive pages and produced a report that can be explained clearly in a portfolio review or interview: performance first, strategic drivers second, and commercial leakage and intervention last.